PAYTON LIAO

+44 7940777287 ◇ London, UK ◇ payton.liao@gmail.com ◇ LinkedIn ◇ GitHub ◇ Portfolio

EDUCATION

MEng Robotics and AI, University College London — Sep 2023 - 2027
Co-founder/Welfare officer- UCL Baseball team

EXPERIENCE

Site Reliability Intern — June 2026 - September 2026
Google — *Dublin, Ireland*

- Built a multi-agent system in Google's internal AI harness to generate configuration files for an internal A/B testing platform, automating SQL queries, configuration syntax, and statistical function selection.
- Developed and evaluated specialised agents using an internal evaluation framework, achieving a 99.8% reduction in configuration creation time.
- Integrated an evaluation pipeline with backtesting on historical evaluations to verify the correctness and sensitivity of generated configurations, adding validation capabilities absent from the previous manual workflow.

Gen-AI Research & Software Engineering Intern — June 2025 - September 2025
MathWorks Generative AI Engineering Team — *Cambridge, UK*

- Designed and deployed a distributed leaderboard platform for benchmarking open-source foundation models at scale, directly informing model selection for fine-tuning MATLAB Copilot used by millions of engineers.
- Built and deployed a full-stack web app (FastAPI, React TSX, MariaDB) with a GPU-accelerated inference pipeline (vLLM) to process and publish evaluation metrics in real time.
- Conducted a comprehensive review of state-of-the-art LLM evaluation methods and leaderboard designs to define project specs, ensuring alignment with real-world decision-making needs.

Computer Vision Research Intern — Apr 2025 - June 2025
UCL Surgical Robotic Vision Group — *London, UK*

- Designed and implemented two deep learning models for hyperspectral image super-resolution (HSI-SR), including a novel variant of a state-of-the-art architecture.
- Curated and preprocessed a 20,000-image custom dataset, and benchmarked models against public HSI datasets, achieving superior performance over existing SOTA baselines.
- Conducted a formal literature review and authored a detailed research report summarizing methodology, results, and future directions.

PROJECTS

Embodied Agent Multimodal Body Schema learning — May 2026
UCL — *London, UK*

- Examined how different toddler intrinsic behaviours impact the quality of body schema learning.

V-Tuber sign language interpreter (first place at AI Hackathon London 2024) — March 2024
Encode Club — *London, UK*

- Won 1st place out of 100+ participants by building a real-time AI V-Tuber that translates speech into sign language using NLP, OpenCV, and MediaPipe.

SKILLS

AI & Machine Learning: Deep Learning, LLMs, Agentic AI, Reinforcement Learning, Computer Vision, Embodied Agents, Tactile Sensing

Programming & Tools: Python, C/C++, PyTorch, OpenCV, Docker, Git, CI/CD, Linux, AWS

Research & Development: Literature Review, Dataset Curation, Experimental Design, Scientific Reporting, Model Evaluation,